

1 Appendix for Section 4

1.1 Growth Impatience Implies Harmenberg Impatience

We show here that **growth impatience** implies the condition imposed by Harmenberg (2021) to guarantee the existence of a permanent income weighted distribution of normalized market resources. Letting f denote the density of the permanent income shock ψ , the impatience condition imposed by Harmenberg (2021) is

$$\log(\mathbf{P}) < \int \log(\mathcal{G}\psi)\psi f(\psi) d\psi. \quad (1)$$

Claim 1. *If growth impatience holds, then (1) holds.*

Proof. Since $\psi \mapsto \log(\mathcal{G}\psi)\psi$ is convex, Jensen's inequality implies

$$\int \log(\mathcal{G}\psi)\psi f(\psi) d\psi \geq \log(\mathcal{G}\mathbb{E}\psi)\mathbb{E}\psi = \log(\mathcal{G}) \quad (2)$$

Since **growth impatience** implies $\mathcal{G} > \mathbf{P}$ and $\log$ is strictly monotone increasing, the result follows. □

1.2 Apparent Balanced Growth in $\bar{c}$ and $\text{cov}(c, \mathbf{p})$

Section 4.2 demonstrates some propositions under the assumption that, when an economy satisfies the **GIC**, there will be constant growth factors $\Omega_{\bar{c}}$ and Ω_{cov} respectively for $\bar{c}$ (the average value of the consumption ratio) and $\text{cov}(c, \mathbf{p})$. In the case of a Szeidl-invariant economy, the main text shows that these are $\Omega_{\bar{c}} = 1$ and $\Omega_{\text{cov}} = \mathcal{G}$. If the economy is Harmenberg- but not Szeidl-invariant, no proof is offered that these growth factors will be constant.

1.3 $\log c$ and $\log \text{cov}(c, \mathbf{p})$ Grow Linearly

Figures 1 and 2 plot the results of simulations of an economy that satisfies Harmenberg- but not Szeidl-invariance with a population of 4 million agents over the last 1000 periods (of a 2000 period simulation).¹ The first figure shows that $\log \bar{c}$ increases apparently linearly. The second figure shows that $\log(-\text{cov}(c, \mathbf{p}))$ also increases apparently linearly. (These results are produced by the notebook `ApndxBalancedGrowthcNrmAndCov.ipynb`).

¹For an exposition of our implementation of Harmenberg's method, see [this supplemental appendix](#).

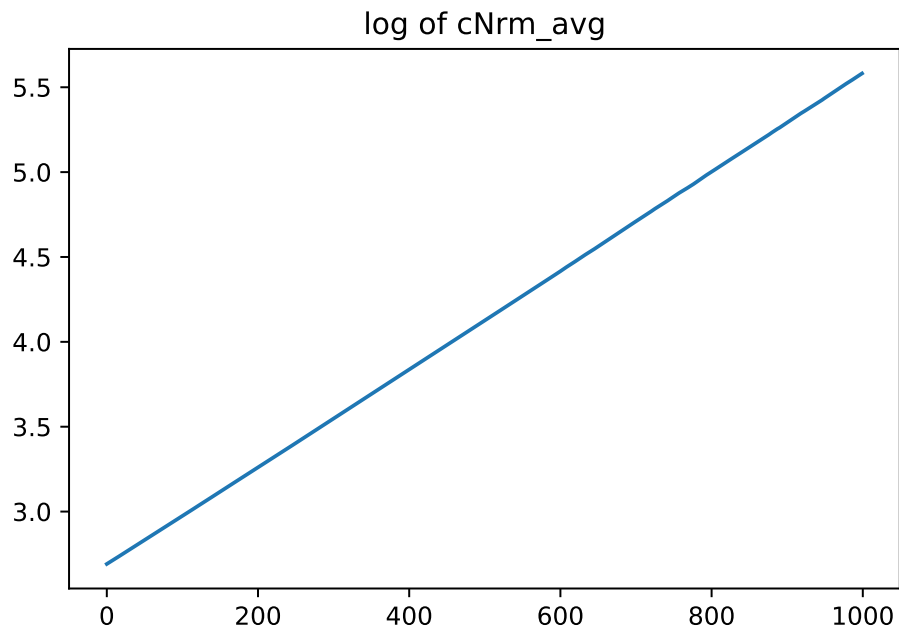


Figure 1 Appendix: $\log \mathfrak{c}$ Appears to Grow Linearly

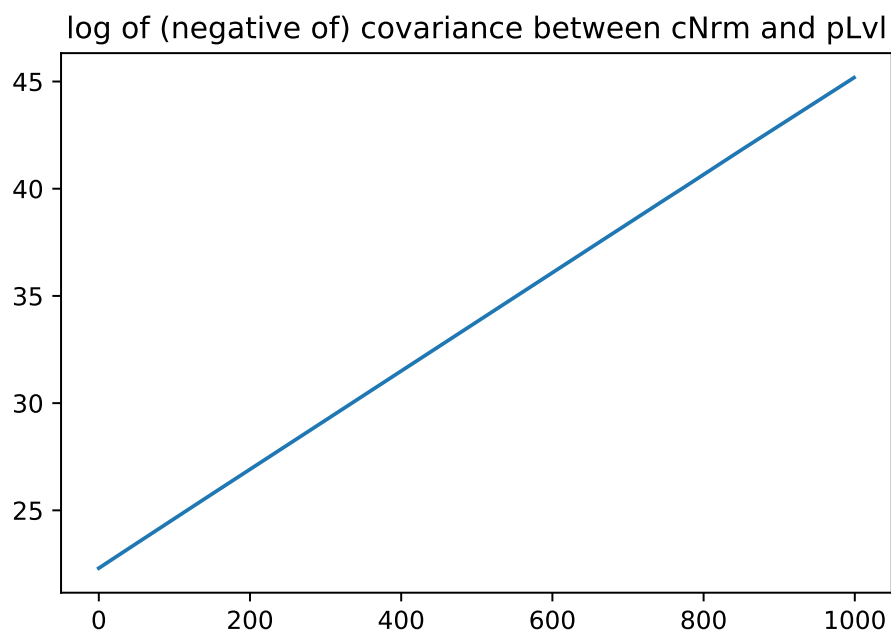


Figure 2 Appendix: $\log (-\text{cov}(c, \boldsymbol{p}))$ Appears to Grow Linearly

(see (2))

References

HARMENBERG, KARL (2021): “Aggregating heterogeneous-agent models with permanent income shocks,” Journal of Economic Dynamics and Control, 129, 104185.